UCLouvain
Institute of Mechanics,
Materials and Civil Engineering

Robust optimisation of the pathway towards a sustainable whole-energy system

A hierarchical multi-objective reinforcement-learning based approach

Doctoral dissertation presented by
Xavier RIXHON

in partial fulfillment of the requirements for
the degree of Doctor in Engineering Sciences

December 2023

Thesis committee
Pr. Francesco CONTINO (supervisor, UCLouvain)
Pr. Hervé JEANMART (supervisor, UCLouvain)
Dr. Stefano MORET (ETH Zurich)
Pr. Sylvain QUOILIN (ULiège)

© 2023
Xavier Rixhon
All Rights Reserved

Abstract

This thesis will be awesome

“Pour ce qui est de l’avenir, il ne s’agit pas de le prévoir,
mais de le rendre possible”

– Antoine de Saint Exupéry, Citadelle, 1948

Remerciements

Thank you, thank you, far too kind

Contents

Symbols **ix**

Introduction **1**

1 Electrofuels and uncertainties in a target future year **3**

1.1 Equilibrium model description 3

1.2 Gasification with varying gasification agents 5

1.2.1 Steam and oxygen 5

1.2.2 Hydrogen and CO_2 5

Conclusions **7**

Bibliography **9**

Symbols

Acronyms

PCE Polynomial Chaos Expansion
UQ Uncertainty Quantification

Introduction

Experimental studies involving simultaneous injection of air, steam and oxygen in two-stage downdraft gasifiers are rare, and often focus on the production of hydrogen rather than a high LHV and low tar content. Furthermore, none of the gasifier designs used in the reviewed studies does include a physical separation of the biomass and char beds and an intermediate oxidation zone: all of them are based on a staged air injection in a continuous porous biomass bed.

With air and saturated steam in a dual-fired downdraft gasifier, **?**] found that a Steam-to-Biomass mass Ratio (SBR) of around 0.2 maximized the syngas H_2 content (27.2%) and Higher Heating Value (HHV) (6.3 MJ/Nm^3). **?**] also investigated air-steam gasification in a two-stage gasifier, and similarly reported that SBR=0.2 yielded more H_2 (up to 16.6%) and a higher HHV (up to 5.3 MJ/Nm^3) compared to the situation with air only. They simultaneously observed an increase in the tar content, explained by the temperature reduction at the location of the steam injection, which negatively affected tar thermal cracking. In these studies, the focus on HHV instead of LHV induces a bias in favour of H_2: water condensation enthalpy is usually not recovered in CHP and industrial burners, and H_2 has a higher HHV but a lower LHV than CO.

Under similar conditions with air and steam in a 50 kW two-stage downdraft gasifier, **?**] found that steam injection resulted in a higher H_2 concentration (up to 20%) but a lower syngas LHV (below 4 MJ/Nm^3), due to the resulting increase in the air-fuel Equivalence Ratio (ER). They also observed that steam injection with air reduced the particle matter concentration (-50% from 102 to 53 mg/Nm^3), but increased the tar content (+69% from 54 to 92 mg/Nm^3) due to a lower temperature inside the gasifier. Using a combination of oxygen and steam, they reported a LHV up to 8.4 MJ/Nm^3 and observed that the use of O_2 mitigated the negative impact of steam on tar content.

?] used oxygen and large amounts of superheated steam (830°C, SBR between 0.5 and 1.7) in a staged open-top gasifier, for hydrogen production. They reported a

syngas LHV up to 8.9 MJ/Nm^3, and observed that as the amount of steam increased, the syngas H_2 content was raised up to 50%, at the expense of a lower syngas LHV.

Only partial agreement was found on the effects of oxygen and steam on syngas LHV and tar content in staged downdraft gasifiers. The diversity of operating conditions, such as the secondary-to-primary O_2 ratio and the Equivalence Ratio (ER), can explain some of the differences observed. Results are often presented in a way that is difficult to generalize to other similar installations. Overall, existing studies show that oxy-steam gasification boosts the syngas LHV, but the individual effects of oxygen and steam are not clearly distinguished as no experiment with oxygen only was found. Steam injection increases the H_2 concentration, but opposite observations exist on its effect on the syngas LHV. Finally, the respective effects of oxygen and steam on tar were not studied in much depth, and seem to depend much on the temperature conditions in the gasifier.

Chapter 1

Electrofuels and uncertainties in a target future year

1.1 Equilibrium model description

Polynomial Chaos Expansion (PCE) i.e. A thermodynamic equilibrium model is used to predict the impact of steam and oxygen injection on the gasification process, and to support the interpretation of experimental results. In this model, the composition of syngas is calculated assuming that all chemical reactions proceed to their equilibrium, disregarding chemical kinetics and process dynamics. [1] Evans et al. [1]

The stoichiometric approach is used for the equilibrium model. The model reactants are: biomass on dry and ash-free basis (daf) represented by the formula $CH_yO_xN_z$, biomass moisture H_2O_l, and the gasification agents N_2, O_2 and steam H_2O_g. At the gasifier outlet, raw syngas is assumed to contain only CO, CO_2, CH_4, H_2, H_2O_g and N_2, and residual solid carbon C_s can subsist in the products if the gasification is incomplete. The 6 unknown coefficients of the products composition (Eq. 1.1) are solved using the elemental mass balances of C, H and O, combined with the equilibrium constants of three reactions. The equilibrium reactions considered are Boudouard reaction (Eq. 1.2), the Water-Gas Shift (WGS) (Eq. 1.3) and steam methane reforming (Eq. 1.4). The equilibrium constants are calculated using NIST-JANAF tables [**?**], for a given equilibrium temperature.

$$\begin{aligned}&\mathrm{CH}_y\mathrm{O}_x\mathrm{N}_z + m\mathrm{H_2O}_l + w\left(\mathrm{O}_2 + \xi\mathrm{N}_2\right) + s\mathrm{H_2O}_g\\&\Rightarrow a_1\mathrm{CO} + a_2\mathrm{CO_2} + a_4\mathrm{CH_4} + b_1\mathrm{H_2} + b_2\mathrm{H_2O}_g + \left(\frac{z}{2} + \xi w\right)\mathrm{N_2} + (a_0 + u_\mathrm{C})\mathrm{C}_s\end{aligned} \tag{1.1}$$

$$\mathrm{C}_s + \mathrm{CO_2} \rightleftharpoons 2\,\mathrm{CO} \tag{1.2}$$

$$\mathrm{CO} + \mathrm{H_2O} \rightleftharpoons \mathrm{CO_2} + \mathrm{H_2} \tag{1.3}$$

$$\mathrm{CH_4} + \mathrm{H_2O} \rightleftharpoons \mathrm{CO} + 3\,\mathrm{H_2} \tag{1.4}$$

The equilibrium temperature is determined, iteratively with the composition of the products, using the energy conservation between the reactants R and the products P (Eq. 1.5). In this equation, $h_i^\circ(T)$ [kJ/kmol] is the total enthalpy (formation and sensible) of species i at temperature T [K] and n_i [kmol/kmol$_{\mathrm{CH}_y\mathrm{O}_x\mathrm{N}_z}$] is the amount of species i in the reactants or products. H_{loss} [kJ/kmol$_{\mathrm{CH}_y\mathrm{O}_x\mathrm{N}_z}$] is an exogenous heat loss, A [kg$_{\mathrm{ash}}$/kmol$_{\mathrm{CH}_y\mathrm{O}_x\mathrm{N}_z}$] is the biomass ash content and $h_A(T_{eq})$ [kJ/kg] is the sensible enthalpy of ash at temperature T [K], determined using a correlation (Eq. 1.6) proposed by **?**].

$$\sum_{i\in R} n_i h_i^\circ(T_{i,0}) = \sum_{i\in P} n_i h_i^\circ(T_{eq}) + A h_A(T_{eq}) + H_{\mathrm{loss}} \tag{1.5}$$

$$h_A(T) = 0.0002155T^2 + 0.7618T - 254.20 \tag{1.6}$$

Steam injection is defined by the Steam-to-Biomass mass ratio (SBR) (Eq. 1.7), which determines the coefficient s, and by the steam temperature T_{st}. Oxygen injection is represented by an equivalent air enrichment affecting the $\mathrm{N_2/O_2}$ ratio, ξ. The amount of oxygen w is defined by the air-fuel Equivalence Ratio (ER) (Eq. 1.8), which is either imposed or computed to maximize the Cold Gas Efficiency (CGE) (Eq. 1.9). Other inputs to the model are the biomass characteristics (elemental composition, LHV, moisture and ash content), the process pressure p (1 bar), the inlet temperature of air and biomass, and the exogenous heat losses H_{loss}.

$$\text{SBR} = \frac{\dot{m}_{\text{H}_2\text{O}_g}}{\dot{m}_{\text{CH}_y\text{O}_x\text{N}_z}} = s\frac{M_{\text{H}_2\text{O}}}{M_{\text{CH}_y\text{O}_x\text{N}_z}} \tag{1.7}$$

$$\text{ER} = \frac{1}{1+\frac{y}{4}-\frac{x}{2}}\frac{\dot{n}_{\text{O}_2}}{\dot{n}_{\text{CH}_y\text{O}_x\text{N}_z}} = \frac{w}{1+\frac{y}{4}-\frac{x}{2}} \tag{1.8}$$

$$\text{CGE} = \frac{P_{\text{syngas}}}{P_{\text{biomass},wb}} = \frac{a_1\text{LHV}_{\text{CO}} + b_1\text{LHV}_{\text{H}_2} + a_4\text{LHV}_{\text{CH}_4}}{\text{LHV}_{\text{CH}_y\text{O}_x\text{N}_z} - mh_{lv,\text{H}_2\text{O}}} \tag{1.9}$$

Two parameters can be adapted to obtain results more representative of real conditions: the exogenous heat losses H_{loss} and the minimum amount of unconverted carbon u_{C}. The amount of unconverted carbon is set to $u_{\text{C}} = 8\%$, based on typical experimental values. This corresponds to the fraction of the carbon that is usually found in the ashes: as char progresses downwards in the reduction zone, it moves towards a lower temperature zone, preventing the complete conversion of carbon. Regarding the heat losses, the best match between the model and the experimental observations is found by setting $H_{\text{loss}} = 2.42\,\text{MJ/kg}_{\text{wood},daf}$.

In the results described in the next section, the Equivalence Ratio (ER) is optimized for each value of the equivalent air enrichment (from air to O_2) without steam. When adding steam, the ER is kept constant to the situation at SBR=0, neglecting any impact of steam on the optimal ER. If the ER is left to be optimized, it rapidly decreases when increasing the SBR, as the added steam theoretically replaces O_2 as gasifying agent. Steam-involved reactions are mainly endothermic, and the equilibrium temperature drops if the model is allowed to seek such optimum: the model then predicts a very high production of CH_4 which is not realistic as the assumption that reactions proceed close to the equilibrium becomes unacceptable. To avoid this effect, the ER was forced not to vary with the SBR.

1.2 Gasification with varying gasification agents

1.2.1 Steam and oxygen

1.2.2 Hydrogen and CO_2

Conclusions

Bibliography

[1] A. Evans, V. Strezov, T. J. Evans, Assessment of utility energy storage options for increased renewable energy penetration, Renewable and Sustainable Energy Reviews 16 (2012) 4141–4147. URL: http://dx.doi.org/10.1016/j.rser.2012.03.048. doi:10.1016/j.rser.2012.03.048.